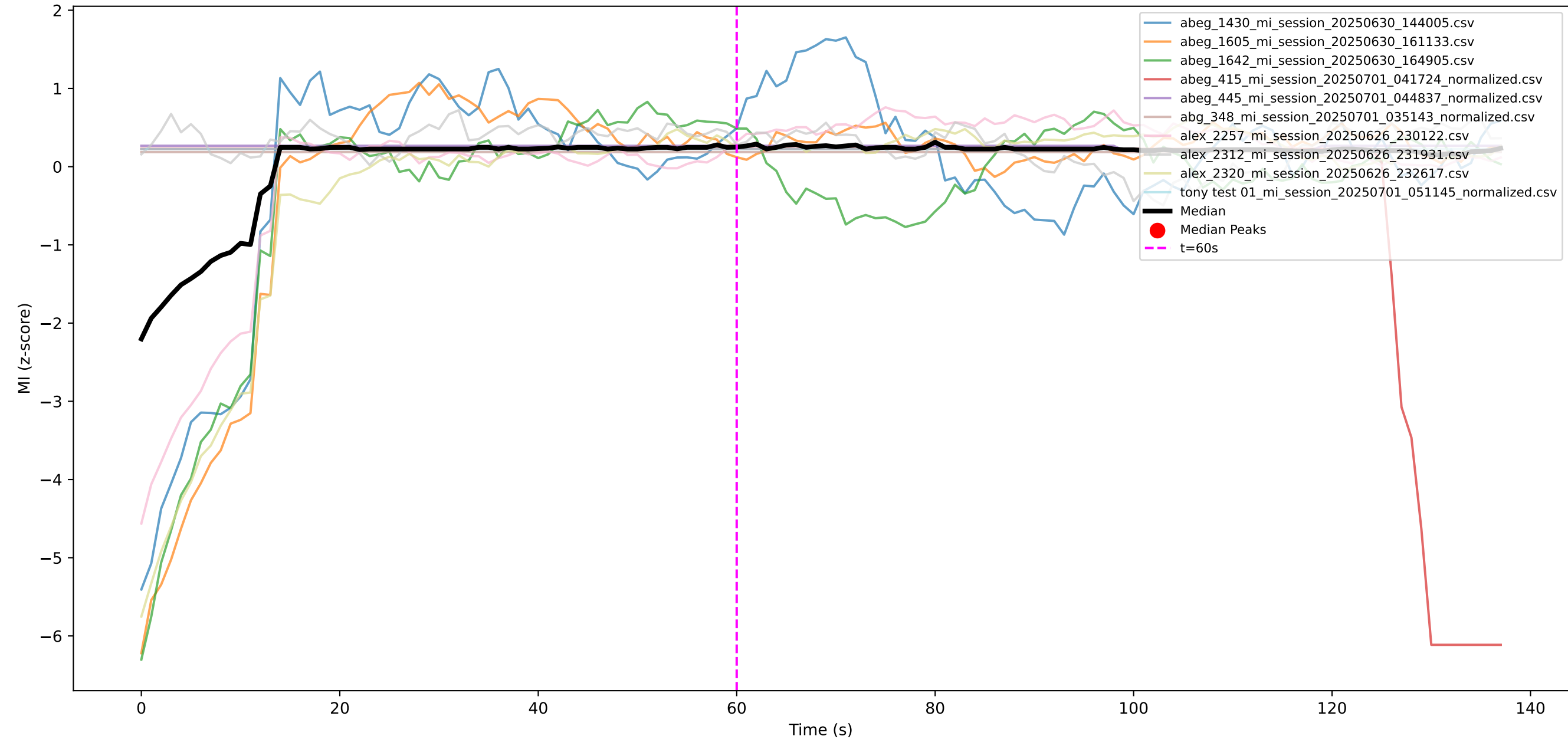
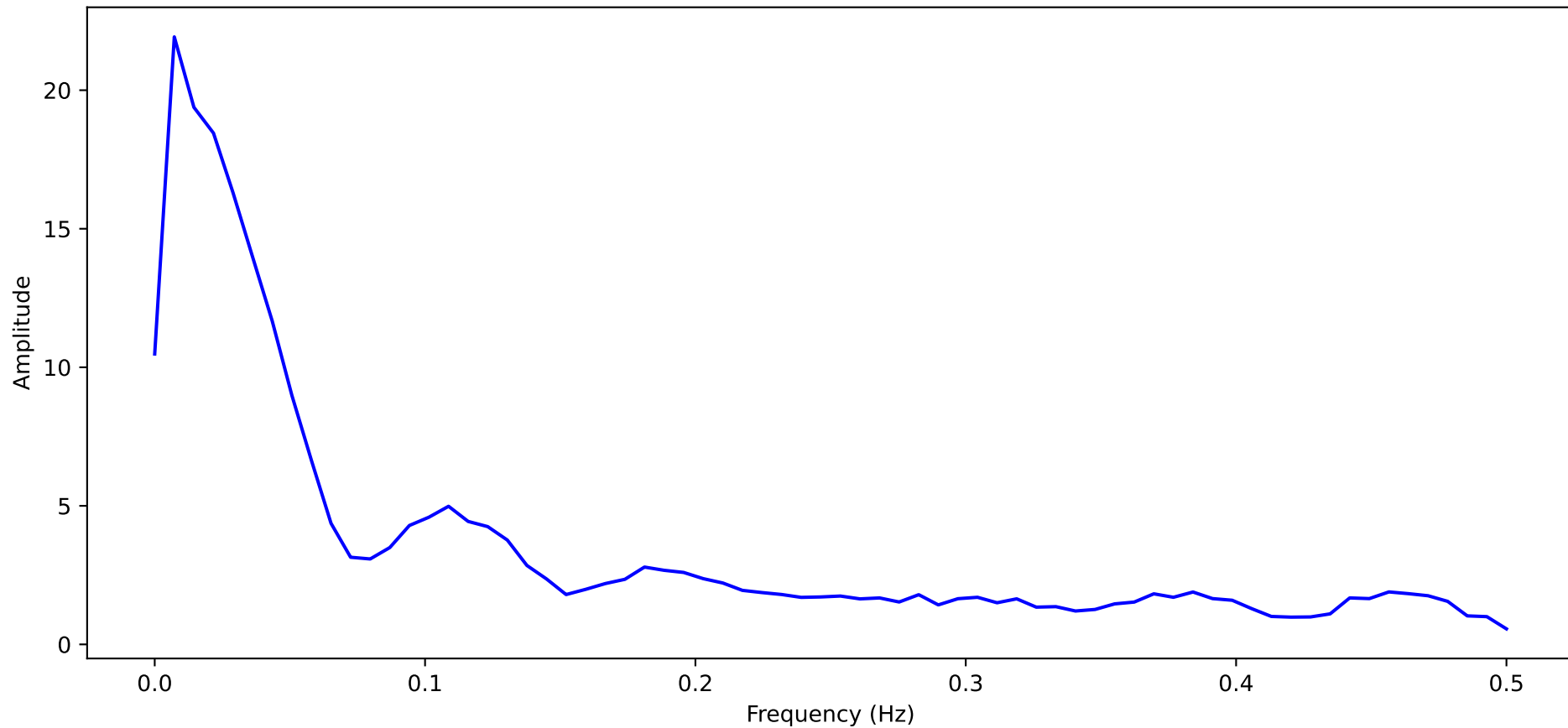


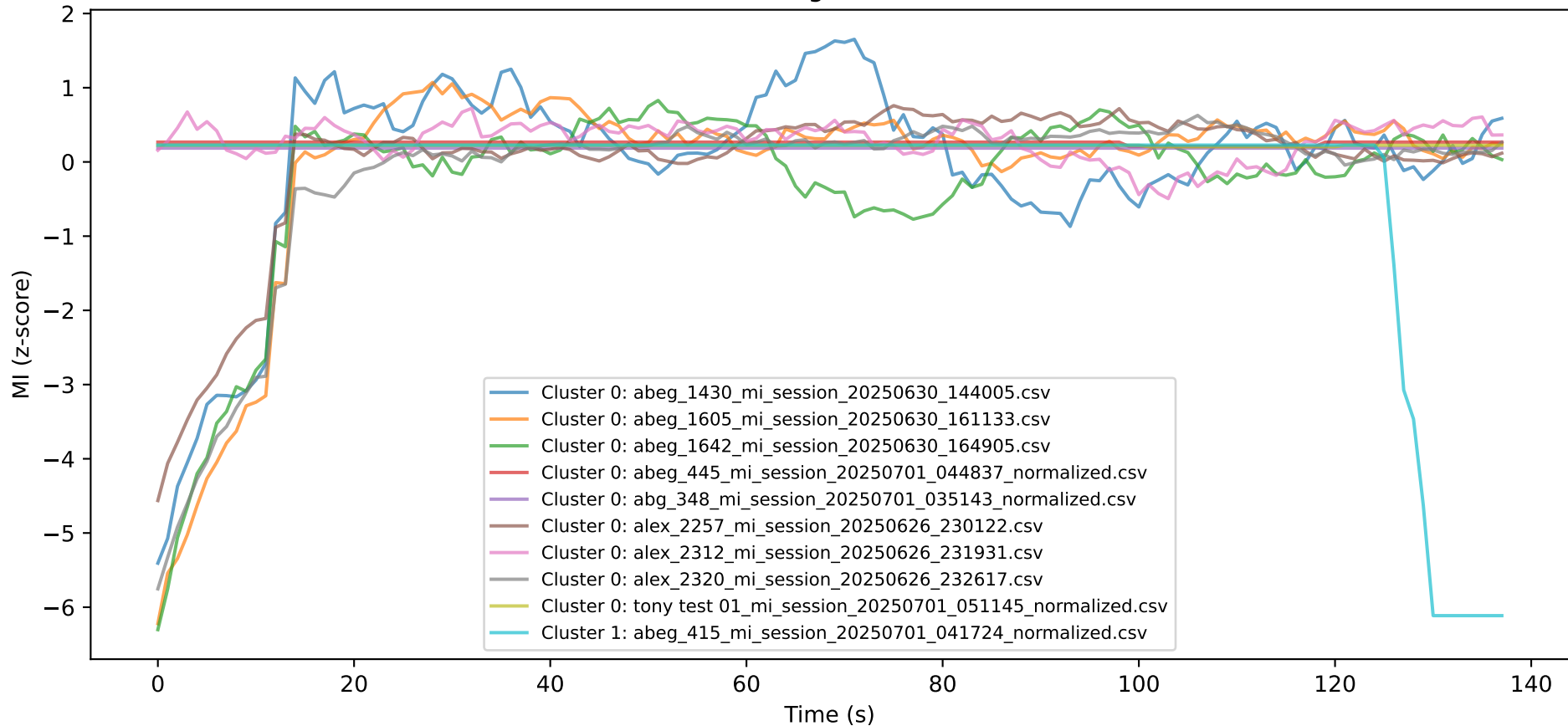
MI Curves (z-score, MA=20) with Median and Peaks



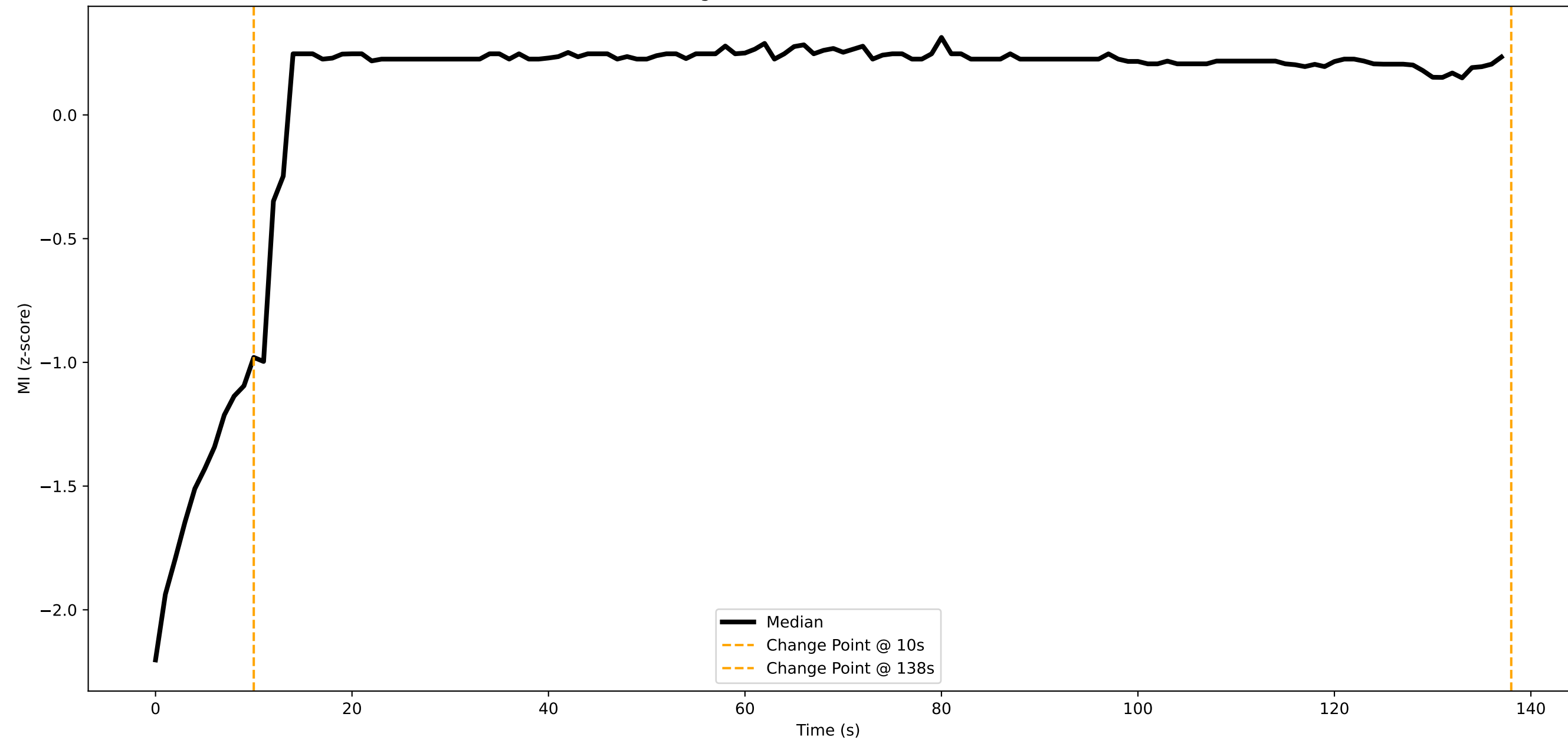
Median MI Frequency Spectrum (FFT)



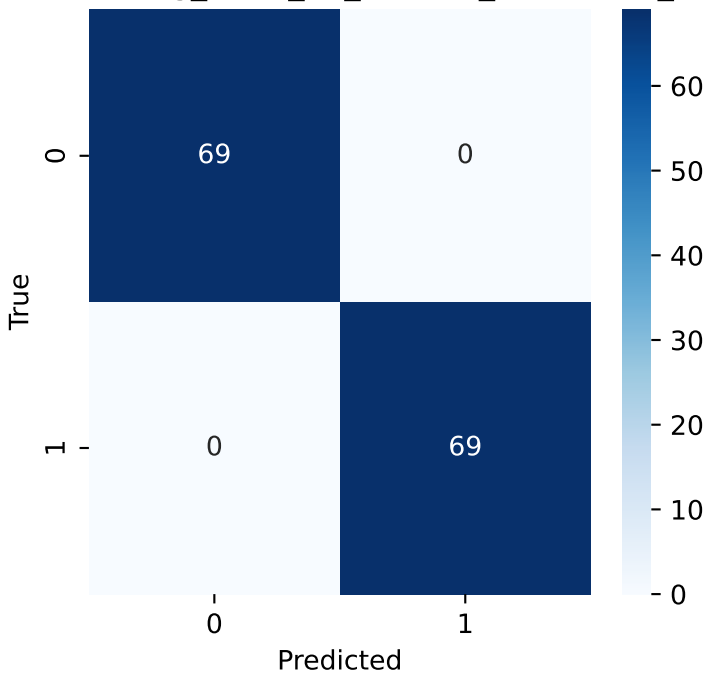
KMeans Clustering of MI Curves (k=2)



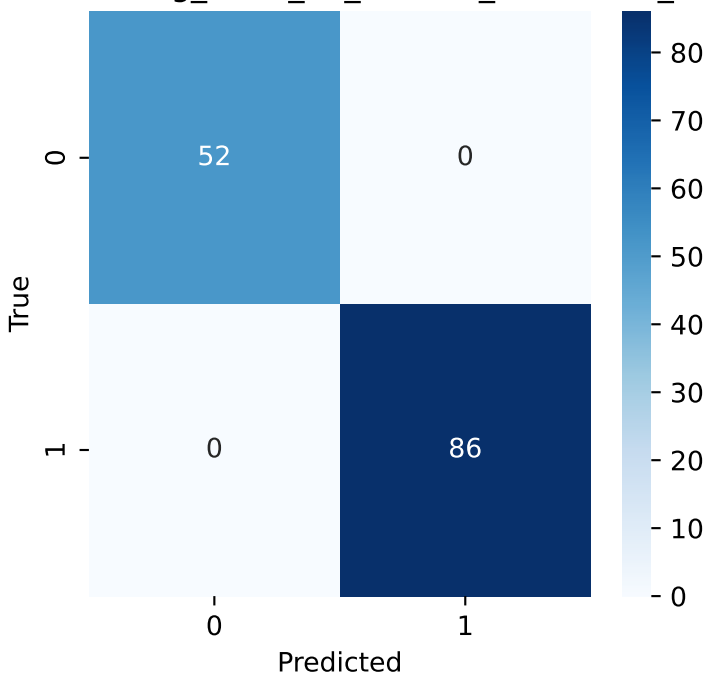
Change Point Detection on Median MI



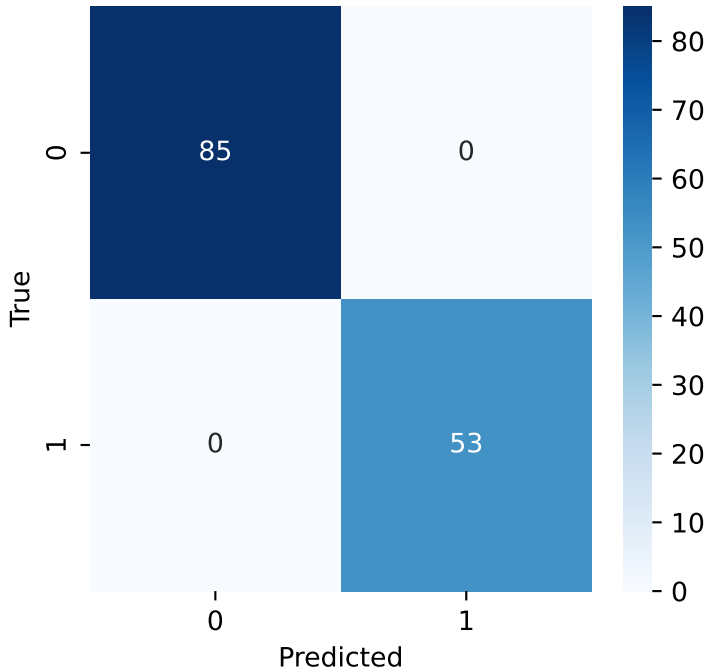
Matrix: abeg_1430_mi_session_20250630_1440



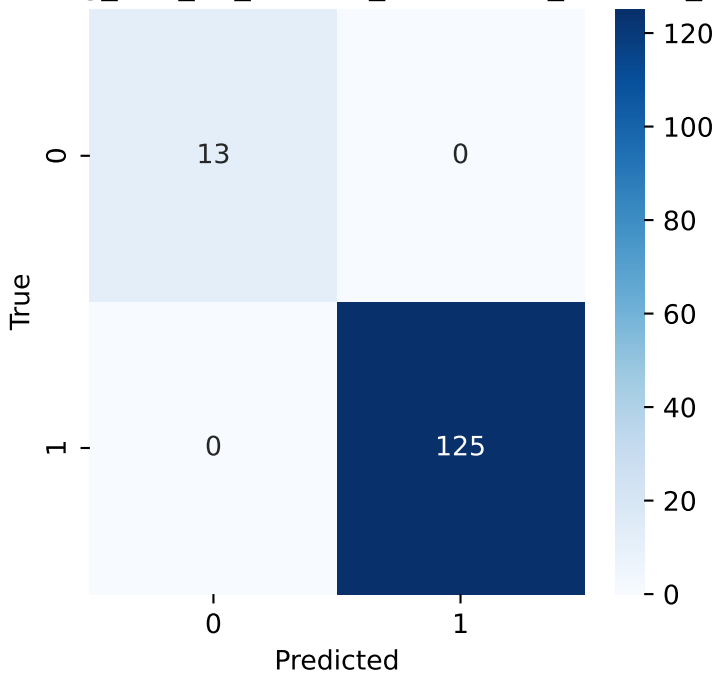
Matrix: abeg_1605_mi_session_20250630_1611



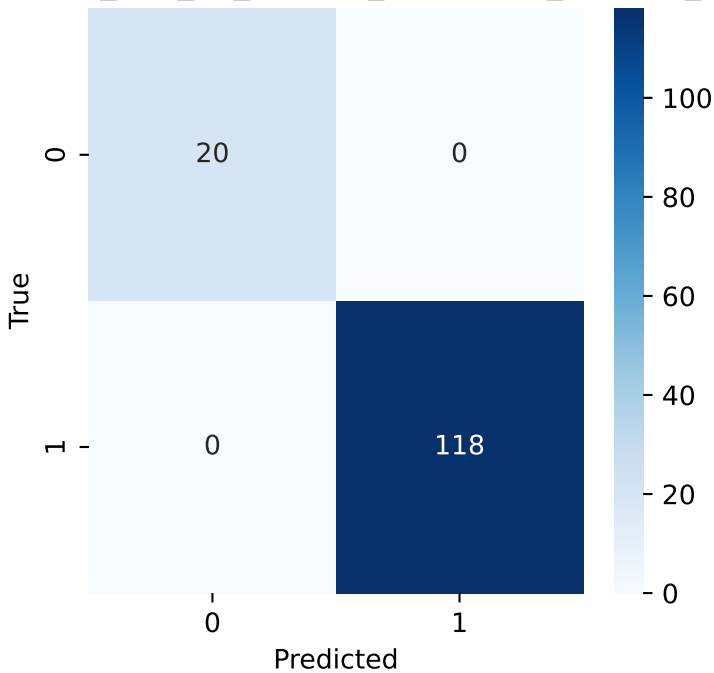
Matrix: abeg_1642_mi_session_20250630_1649



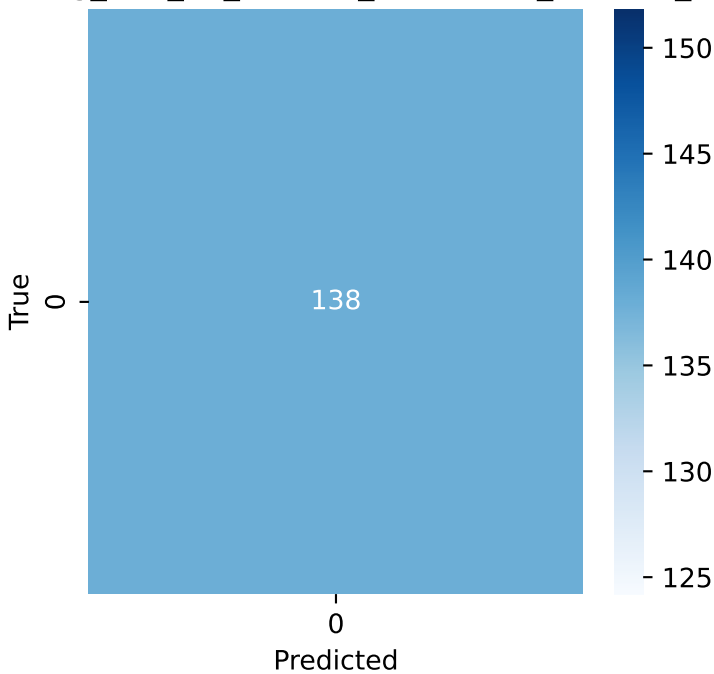
x: abeg_415_mi_session_20250701_041724_no



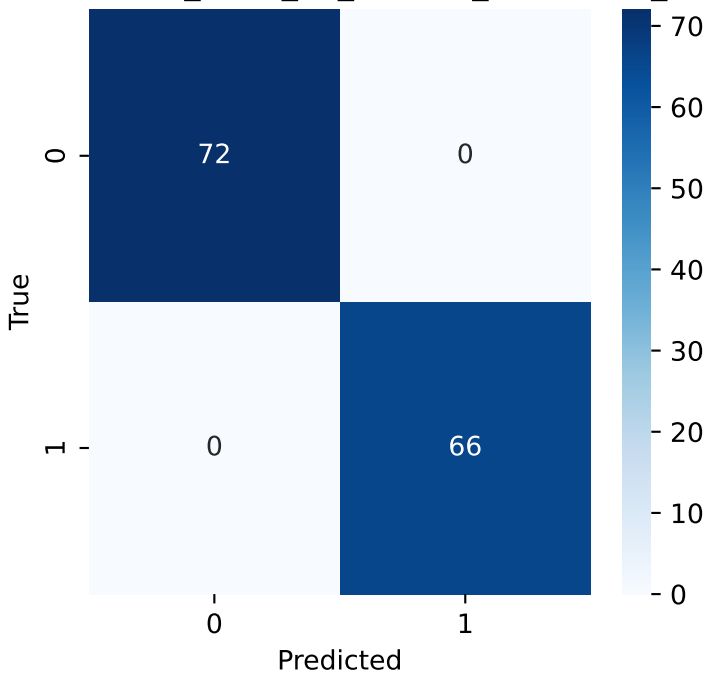
x: abeg_445_mi_session_20250701_044837_no



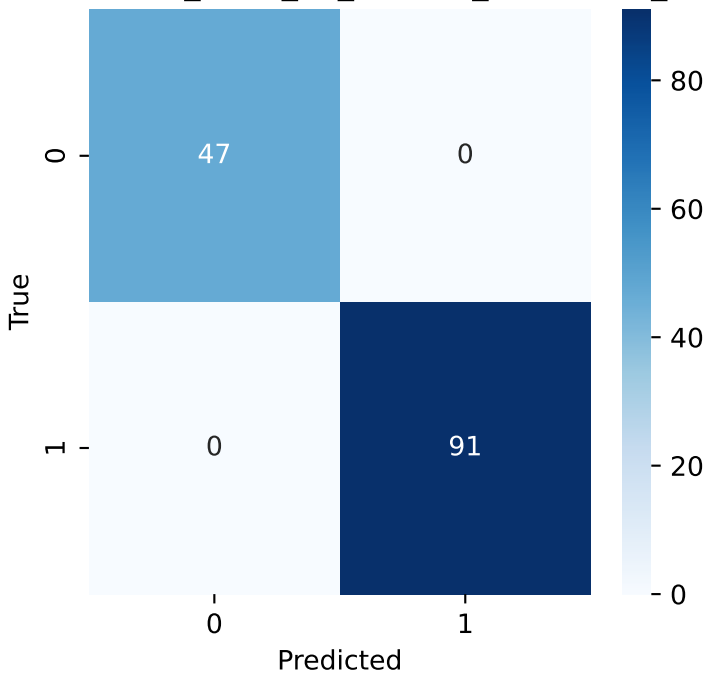
ix: abg_348_mi_session_20250701_035143_nor



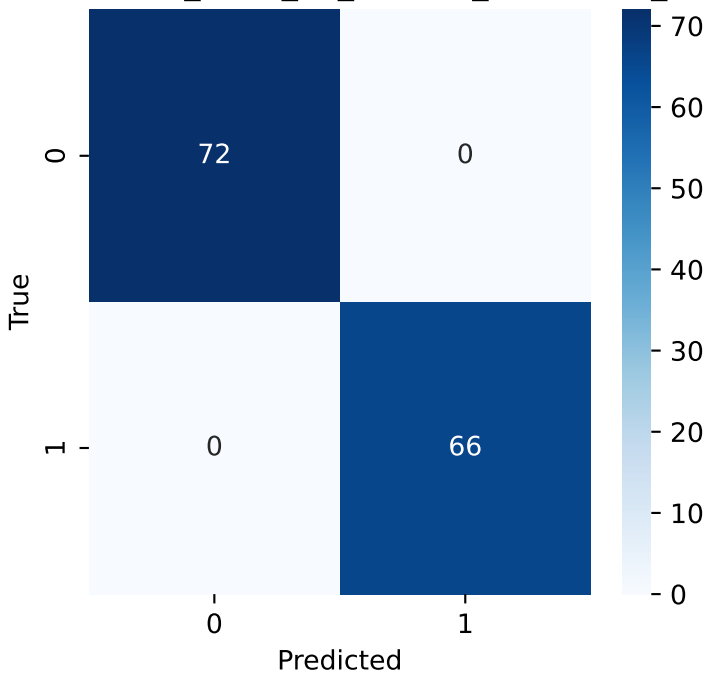
h Matrix: alex_2257_mi_session_20250626_2301



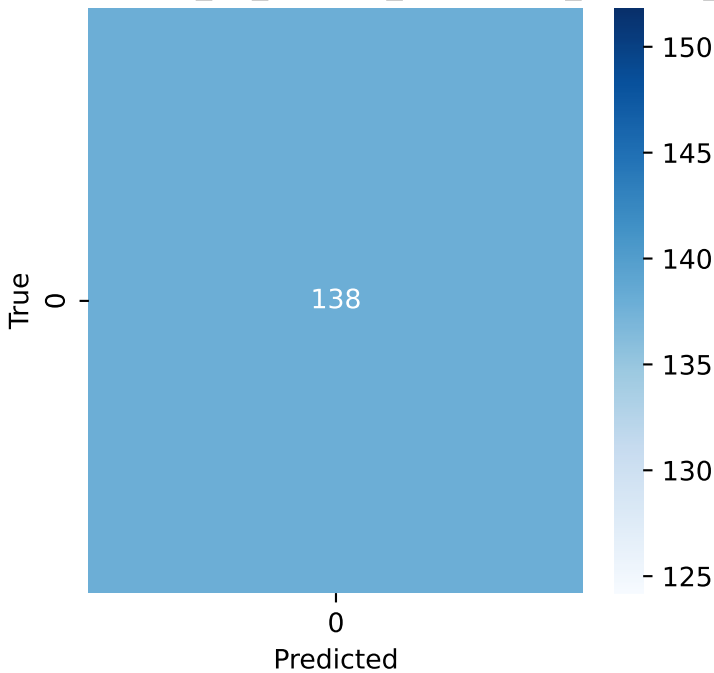
h Matrix: alex_2312_mi_session_20250626_2319



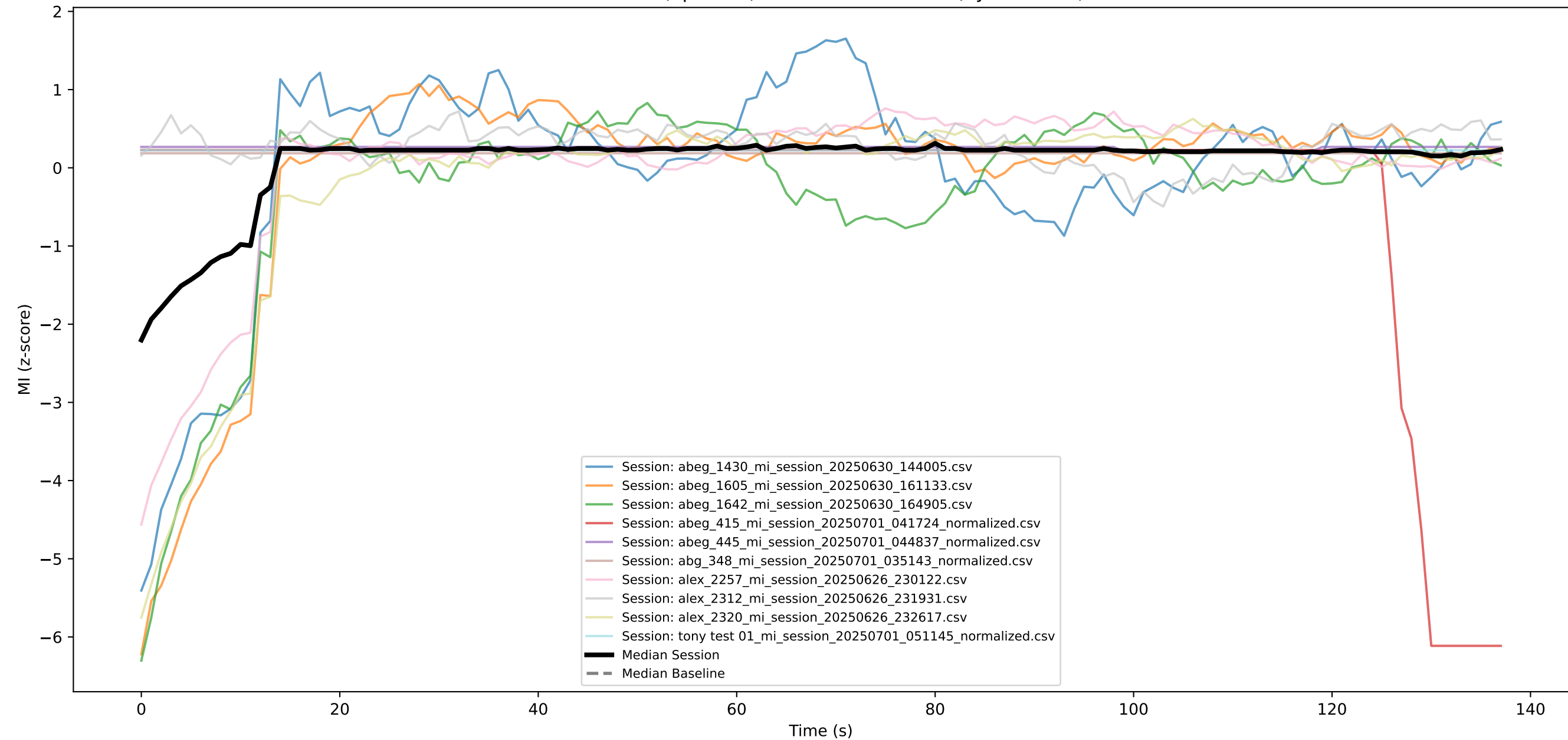
h Matrix: alex_2320_mi_session_20250626_2326



: tony test 01_mi_session_20250701_051145_no



Session MI (up to 10) vs. Median Baseline (Eyes Closed)



MI Advanced Report

Generated: 20250717_022951

MI Advanced Report - Statistical Summary

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 100 samples were included.
- * Individual and median MI curves were plotted (see Figure 8.1).
- * RM-ANOVA across five epochs (0-30s, 31-90s, 91-150s, 151-240s, 241-300s):

Anova

```
=====
F Value Num DF Den DF Pr > F
-----
Epoch    nan 4.0000 36.0000    nan
=====
```

- * Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1 group2 meandiff p-adj lower upper reject
-----
0-30s 151-240s    nan    nan    nan    nan False
0-30s 241-300s    nan    nan    nan    nan False
0-30s 31-90s  0.8523    nan    nan    nan False
0-30s 91-150s  0.6301    nan    nan    nan False
151-240s 241-300s    nan    nan    nan    nan False
151-240s 31-90s    nan    nan    nan    nan False
151-240s 91-150s    nan    nan    nan    nan False
241-300s 31-90s    nan    nan    nan    nan False
241-300s 91-150s    nan    nan    nan    nan False
31-90s 91-150s -0.2222    nan    nan    nan False
=====
```

- * Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at times [], amplitudes [], durations [] (see Table 8.2, Figure 8.2).
- * AUC (median): 11.46; mean $\pm$ SD (individual): 4.31 $\pm$ 25.87 (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-150s vs 151-300s): mean diff = 0.000, CI = [0.000, 0.000].
- * Change point detection (Pelt algorithm): 1 change points at [10] (see Figure 8.5).

3. Clustering & Mixed-Effects

- * K-Means clustering ($k=2$): cluster averages {}, user distribution {0: 9, 1: 1} (see Figure 8.6).
- * Mixed-effects model (MI $\sim$ time + (1|Subject)): Mixed Linear Model Regression Results

```
=====
Model:          MixedLM Dependent Variable: value
No. Observations: 1380 Method:          REML
No. Groups:      10 Scale:              1.0048
Min. group size: 138 Log-Likelihood:    -1977.5717
Max. group size: 138 Converged:         Yes
Mean group size: 138.0
=====
```

```
-----
Coef. Std.Err. z P>|z| [0.025 0.975]
-----
Intercept -0.307 0.081 -3.815 0.000 -0.465 -0.149
time       0.005 0.001 7.056 0.000 0.003 0.006
file Var   0.036 0.020
=====
```

- * ANOVA on final epoch by cluster: [ANOVA final epoch analysis not available] (see Table 8.5).

4. EDA & Correlations

- * EDA RM-ANOVA: [EDA RM-ANOVA results not available]